

Characterization and Genomic Analysis of the Lyme disease spirochete bacteriophage phiBB-1 Fig 5B, 5C, and 5D.

phiBB-1 virions package cp32 isoforms, chromosome fragments, lp54, and other plasmid
DNA.

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The following analysis represents an independent reconstruction of Figures 5B, 5C, and 5D from the listed publication: Faith DR, Kinnersley M, Brooks DM, Drecktrah D, Hall LS, Luo E, et al. (2024) Characterization and genomic analysis of the Lyme disease spirochete bacteriophage phiBB-1. PLoS Pathog 20(4): e1012122. <https://doi.org/10.1371/journal.ppat.1012122>

Publicly available sequencing data found under the journal article was used to generate tables that were then visualized using ggplot2 to replicate the original figures.

This was done as a final bioinformatics project for an independent study at the Qiu lab at CUNY Hunter College for a Biology MA.

Sequence alignment and filtering according to the methods of the publication can be found in the seqpipe.sh script.

```
setwd("/Users/ashqulislam/Desktop/QiuLab/BB1 Figure Project/")
library(tidyverse)
library(knitr)
library(kableExtra)
#Open reference, idx, and coverage files
ref <- read_tsv("reference_lookup_tabs.tsv",
               col_names = c("Reference", "Replicon"))

kable(ref,
      caption = "Reference Table",
      booktabs = TRUE)
```

Table 1: Reference Table

Reference	Replicon
NC_012157.1	cp26
NC_012158.1	cp32-1
NC_012152.1	cp32-10
NC_012156.1	cp32-13
NC_012154.1	cp32-3
NC_012149.1	cp32-5
NC_012227.1	lp17
NC_012228.1	lp28-3

Reference	Replicon
NC_012202.1	lp36-28-4
NC_012170.1	lp38
NC_012194.1	lp54
NC_012203.1	lp56
NZ_ABJY02000014.1	whole
NZ_ABJY02000013.1	whole
NZ_ABJY02000012.1	whole
NZ_ABJY02000011.1	whole
NZ_ABJY02000010.1	whole
NZ_ABJY02000009.1	whole
NZ_ABJY02000008.1	whole
NZ_ABJY02000007.1	whole
NZ_ABJY02000006.1	whole
NZ_ABJY02000005.1	whole
NZ_ABJY02000004.1	whole
NZ_ABJY02000003.1	whole
NZ_ABJY02000002.1	whole
NZ_ABJY02000001.1	whole

```
aln1f_idx <- read_tsv("aln1_filtered_idxstats.tsv",
                     col_names = c("Reference",
                                   "Length",
                                   "Mapped_Reads",
                                   "Unmapped_Reads"))

#Removed an unknown tab that wasn't tied to Chromosome or plasmids
#Aligned the reference_lookup tsv which had the replicon names aligned to Reference
aln1f_idx <- aln1f_idx |>
  filter (Reference != "*") |>
  left_join(ref, by = "Reference")

# Rearranged rows to account for left_join
aln1f_idx <- aln1f_idx |>
  select(Reference, Replicon, everything())
#The Chromosome section had multiple rows, assumed it took several reads to
#cover it. whole_row1 is the isolation of the rows via filter and then summarize
#to condense values. Due to multiple rows encompassing one variable, values are summed.

#ref table and tsv's lists chromosome as "whole".
#Renamed to "chromosome" for readability

whole_row1f <- aln1f_idx |>
  filter(Replicon == "whole") |>
  summarize(
    Reference = "Chromosome",
    Length = sum(Length),
    Mapped_Reads = sum(Mapped_Reads),
    Unmapped_Reads = sum(Unmapped_Reads),
    Replicon = "chromosome")

#Bind the row back to aln1_idx
```

```
aln1f_idx <- aln1f_idx |>
  filter(Replicon != "whole") |>
  bind_rows(whole_row1f)

kable(aln1f_idx,
  caption = "Idxstats table following modifications.",
  booktabs = TRUE)
```

Table 2: Idxstats table following modifications.

Reference	Replicon	Length	Mapped_Reads	Unmapped_Reads
NC_012157.1	cp26	26511	14	0
NC_012158.1	cp32-1	30408	28	0
NC_012152.1	cp32-10	30290	1956	0
NC_012156.1	cp32-13	30088	761	0
NC_012154.1	cp32-3	29762	1647	0
NC_012149.1	cp32-5	29912	908	0
NC_012227.1	lp17	18181	5	0
NC_012228.1	lp28-3	25799	1	0
NC_012202.1	lp36-28-4	54623	11	0
NC_012170.1	lp38	27775	7	0
NC_012194.1	lp54	54021	208	0
NC_012203.1	lp56	29418	17	0
Chromosome	chromosome	907566	216	0

*#Coverage file needed a rename. Followed by a join of ref tsv along with
#reordering the columns*

```
aln1f_cov <- read_tsv("aln1_filtered_coverage.tsv",
  comment = "")

kable(aln1f_cov,
  caption = "Coverage Table 1 Unedited",
  booktabs = TRUE)
```

Table 3: Coverage Table 1 Unedited

#rname	startpos	endpos	numreads	covbases	coverage	meandepth	meanbaseq	meanmapq
NC_012157.1	1	26511	14	24537	92.5540	2.843760	29.2	60.0
NC_012158.1	1	30408	28	30408	100.0000	11.039600	35.1	59.8
NC_012152.1	1	30290	1956	30290	100.0000	443.279000	34.0	60.0
NC_012156.1	1	30088	761	30088	100.0000	339.719000	34.0	58.7
NC_012154.1	1	29762	1647	29762	100.0000	739.922000	33.9	58.9
NC_012149.1	1	29912	908	29912	100.0000	383.699000	34.3	59.9
NC_012227.1	1	18181	5	12961	71.2887	1.284970	36.8	60.0
NC_012228.1	1	25799	1	5326	20.6442	0.206442	34.7	60.0
NC_012202.1	1	54623	11	39221	71.8031	1.089190	33.4	60.0
NC_012170.1	1	27775	7	20689	74.4878	1.635100	32.5	60.0
NC_012194.1	1	54021	208	54021	100.0000	21.704400	34.2	59.9
NC_012203.1	1	29418	17	29051	98.7525	3.111050	27.1	60.0

#rname	startpos	endpos	numreads	covbases	coverage	meandepth	meanbaseq	meanmapq
NZ_ABJY02000014.1	1	25946	3	13498	52.0234	0.608071	36.0	60.0
NZ_ABJY02000013.1	1	57720	16	51133	88.5880	1.798320	35.8	60.0
NZ_ABJY02000012.1	1	83077	19	72459	87.2191	1.484770	30.3	60.0
NZ_ABJY02000011.1	1	84891	32	83613	98.4945	2.520410	34.5	60.0
NZ_ABJY02000008.1	1	209169	49	160745	76.8493	1.514570	34.6	60.0
NZ_ABJY02000007.1	1	200423	37	153342	76.5092	1.173730	34.3	60.0
NZ_ABJY02000006.1	1	29093	9	27643	95.0160	1.539610	31.7	60.0
NZ_ABJY02000003.1	1	59106	14	51326	86.8372	1.660980	28.4	60.0
NZ_ABJY02000002.1	1	150291	36	110062	73.2326	1.410350	34.7	60.0
NZ_ABJY02000001.1	1	1206	1	1203	99.7512	0.997512	39.5	60.0
NZ_ABJY02000010.1	1	2419	0	0	0.0000	0.000000	0.0	0.0
NZ_ABJY02000009.1	1	1414	0	0	0.0000	0.000000	0.0	0.0
NZ_ABJY02000005.1	1	1406	0	0	0.0000	0.000000	0.0	0.0
NZ_ABJY02000004.1	1	1405	0	0	0.0000	0.000000	0.0	0.0

```
aln1f_cov<-aln1f_cov |>
  rename(Reference = '#rname')
aln1f_cov <- aln1f_cov |>
  left_join(ref, by = "Reference")|>
  select(Reference, Replicon, everything())

#Chromosome row summarize again but for coverage file. Temporary copies of endpos
#and numreads were used to prevent errors during weights.
#Weights could conflict with sum calculations due to length mismatch.
#Weighted means in respect to numreads to account for variability

whole_row11f <- aln1f_cov |>
  filter(Replicon == "whole")|>
  mutate(
    fragment_length = endpos,
    fragment_reads = numreads
  )

whole_row11f <- whole_row11f |>
  summarize(
    Reference = "Chromosome",
    Replicon = "chromosome",
    startpos = mean(startpos),
    endpos = sum(endpos),
    numreads = sum(numreads),
    covbases = sum(covbases),
    coverage = weighted.mean(x =coverage,
                             w = fragment_length),
    meandepth = weighted.mean(x = meandepth,
                              w = fragment_length),
    meanbaseq = weighted.mean(x = meanbaseq,
                              w = fragment_reads),
    meanmapq = weighted.mean(x = meanmapq,
                              w = fragment_reads)
  )
aln1f_cov <- aln1f_cov |>
```

```

filter(Replicon != "whole")|>
bind_rows(whole_row11f)

#coverage table with relevant columns, ref tabled joined in, summarized chromosome

aln1f_cov |>
select(Reference, Replicon, endpos, numreads, coverage, meandepth, meanmapq)|>
kable(
  caption = "Replicate 1 coverage table post-changes.",
  booktabs = TRUE)

```

Table 4: Replicate 1 coverage table post-changes.

Reference	Replicon	endpos	numreads	coverage	meandepth	meanmapq
NC_012157.1	cp26	26511	14	92.55400	2.843760	60.0
NC_012158.1	cp32-1	30408	28	100.00000	11.039600	59.8
NC_012152.1	cp32-10	30290	1956	100.00000	443.279000	60.0
NC_012156.1	cp32-13	30088	761	100.00000	339.719000	58.7
NC_012154.1	cp32-3	29762	1647	100.00000	739.922000	58.9
NC_012149.1	cp32-5	29912	908	100.00000	383.699000	59.9
NC_012227.1	lp17	18181	5	71.28870	1.284970	60.0
NC_012228.1	lp28-3	25799	1	20.64420	0.206442	60.0
NC_012202.1	lp36-28-4	54623	11	71.80310	1.089190	60.0
NC_012170.1	lp38	27775	7	74.48780	1.635100	60.0
NC_012194.1	lp54	54021	208	100.00000	21.704400	59.9
NC_012203.1	lp56	29418	17	98.75250	3.111050	60.0
Chromosome	chromosome	907566	216	79.88663	1.504091	60.0

This process is replicated for tsvs 2 and 3. A final coverage table will be included for visualization but the code will be hidden during Knit.

Table 5: Replicate 2 coverage table post-changes.

Reference	Replicon	endpos	numreads	coverage	meandepth	meanmapq
NC_012157.1	cp26	26511	107	100.00000	23.119100	60.0
NC_012158.1	cp32-1	30408	208	100.00000	73.250400	58.9
NC_012152.1	cp32-10	30290	10452	100.00000	2463.880000	60.0
NC_012156.1	cp32-13	30088	3069	100.00000	1690.250000	59.2
NC_012154.1	cp32-3	29762	7908	100.00000	4396.900000	59.1
NC_012149.1	cp32-5	29912	3875	100.00000	2111.950000	59.9
NC_012227.1	lp17	18181	33	100.00000	10.977300	60.0
NC_012228.1	lp28-3	25799	3	49.57940	0.679212	60.0
NC_012202.1	lp36-28-4	54623	81	93.62720	10.378200	60.0
NC_012170.1	lp38	27775	52	100.00000	11.973400	60.0
NC_012194.1	lp54	54021	1232	100.00000	136.835000	60.0
NC_012203.1	lp56	29418	58	99.99320	14.279900	60.0
Chromosome	chromosome	907566	1451	99.13483	10.507722	60.0

Table 6: Replicate 3 coverage table post-changes.

Reference	Replicon	endpos	numreads	coverage	meandepth	meanmapq
NC_012157.1	cp26	26511	12	99.9925	2.620690	60.0000
NC_012158.1	cp32-1	30408	67	100.0000	23.914100	59.9000
NC_012152.1	cp32-10	30290	10204	100.0000	2214.760000	60.0000
NC_012156.1	cp32-13	30088	3796	100.0000	2246.040000	59.0000
NC_012154.1	cp32-3	29762	7518	100.0000	4386.070000	59.1000
NC_012149.1	cp32-5	29912	1787	100.0000	1033.700000	60.0000
NC_012227.1	lp17	18181	3	58.1761	1.226450	60.0000
NC_012202.1	lp36-28-4	54623	5	35.3020	0.635172	60.0000
NC_012170.1	lp38	27775	6	69.6958	1.512400	60.0000
NC_012194.1	lp54	54021	145	100.0000	16.356800	60.0000
NC_012203.1	lp56	29418	6	68.1182	1.065670	60.0000
NC_012228.1	lp28-3	25799	0	0.0000	0.000000	0.0000
Chromosome	chromosome	907566	168	62.1535	1.222793	59.8631

```

#Combine all coverage tables, include replicate numbers

aln1f_cov <- aln1f_cov |>
  mutate(Replicate = "Replicate 1")
aln2f_cov <- aln2f_cov |>
  mutate(Replicate = "Replicate 2")
aln3f_cov <- aln3f_cov |>
  mutate(Replicate = "Replicate 3")
cov_allf <- bind_rows (aln1f_cov, aln2f_cov, aln3f_cov)

#Calculate percent reads for Figure 5B and remove grouping

cov_allf <- cov_allf |>
  group_by(Replicate) |>
  mutate(percent_reads = numreads / sum(numreads) *100) |>
  ungroup()

#An arrangement of replicon order to match publication figures

replicon_order <- c("cp32-1", "cp32-3", "cp32-5", "cp32-10", "cp32-13",
  "chromosome", "lp54", "cp26", "lp17", "lp36-28-4", "lp38",
  "lp56", "lp28-3")

```

Table 7: Shortened Combined Coverage Table

Reference	Replicon	endpos	numreads	coverage	meandepth	meanmapq	Replicate
NC_012157.1	cp26	26511	14	92.55400	2.843760	60.0	Replicate 1
NC_012158.1	cp32-1	30408	28	100.00000	11.039600	59.8	Replicate 1
NC_012152.1	cp32-10	30290	1956	100.00000	443.279000	60.0	Replicate 1
NC_012156.1	cp32-13	30088	761	100.00000	339.719000	58.7	Replicate 1
NC_012154.1	cp32-3	29762	1647	100.00000	739.922000	58.9	Replicate 1
NC_012149.1	cp32-5	29912	908	100.00000	383.699000	59.9	Replicate 1
NC_012227.1	lp17	18181	5	71.28870	1.284970	60.0	Replicate 1
NC_012228.1	lp28-3	25799	1	20.64420	0.206442	60.0	Replicate 1

Reference	Replicon	endpos	numreads	coverage	meandepth	meanmapq	Replicate
NC_012202.1	lp36-28-4	54623	11	71.80310	1.089190	60.0	Replicate 1
NC_012170.1	lp38	27775	7	74.48780	1.635100	60.0	Replicate 1
NC_012194.1	lp54	54021	208	100.00000	21.704400	59.9	Replicate 1
NC_012203.1	lp56	29418	17	98.75250	3.111050	60.0	Replicate 1
Chromosome	chromosome	907566	216	79.88663	1.504091	60.0	Replicate 1
NC_012157.1	cp26	26511	107	100.00000	23.119100	60.0	Replicate 2
NC_012158.1	cp32-1	30408	208	100.00000	73.250400	58.9	Replicate 2

```

#calculate a mean percent read and se based off all replicates

figBf_summary <- cov_allf |>
  group_by(Replicon) |>
  summarize(
    mean_percent = mean(percent_reads),
    se_percent = sd(percent_reads)/sqrt(n()),
    .groups = "drop"
  ) |>
  mutate(Replicon = factor(Replicon, levels = replicon_order))

#setup to include replicates

cov_all_plot <- cov_allf |>
  mutate(
    Replicate = factor(
      Replicate,
      levels = c("Replicate 1", "Replicate 2", "Replicate 3")
    ),
    Replicon = factor(Replicon, levels = replicon_order)
  )

figB <- ggplot(data = figBf_summary)+
  aes(x = Replicon, y = mean_percent)+
  coord_cartesian(xlim = NULL, ylim = c(0,45))+
  labs(x = NULL, y = "% reads")+
  theme_classic(base_size = 18)+
  scale_y_continuous(expand = c(0,0))+ #remove offset on x-axis
  theme(
    axis.line = element_line(linewidth = 0.8),
    axis.ticks = element_line(linewidth = 1.2),
    axis.ticks.length = unit(0.2, "cm"),
    axis.text.x = element_text(angle = 55, hjust =1, size = 16)
  )+
  geom_col(
    fill = "royalblue4",
    color = "black",
    linewidth = 0.8
  ) +
  geom_errorbar(
    aes(
      ymin = mean_percent - se_percent,
      ymax = mean_percent + se_percent)))+
  geom_point(

```

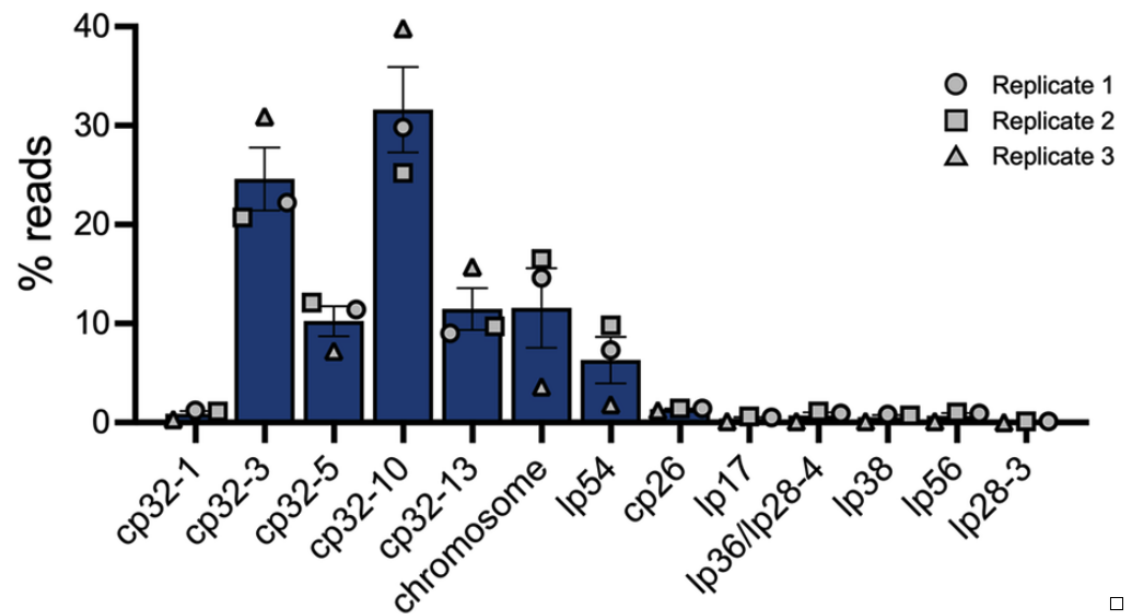
```

data = cov_all_plot,
aes(x = Replicon, y = percent_reads, shape = Replicate),
position = position_jitterdodge(jitter.width = 0.05,
                                dodge.width = 0.5), #prevent replicate overlap

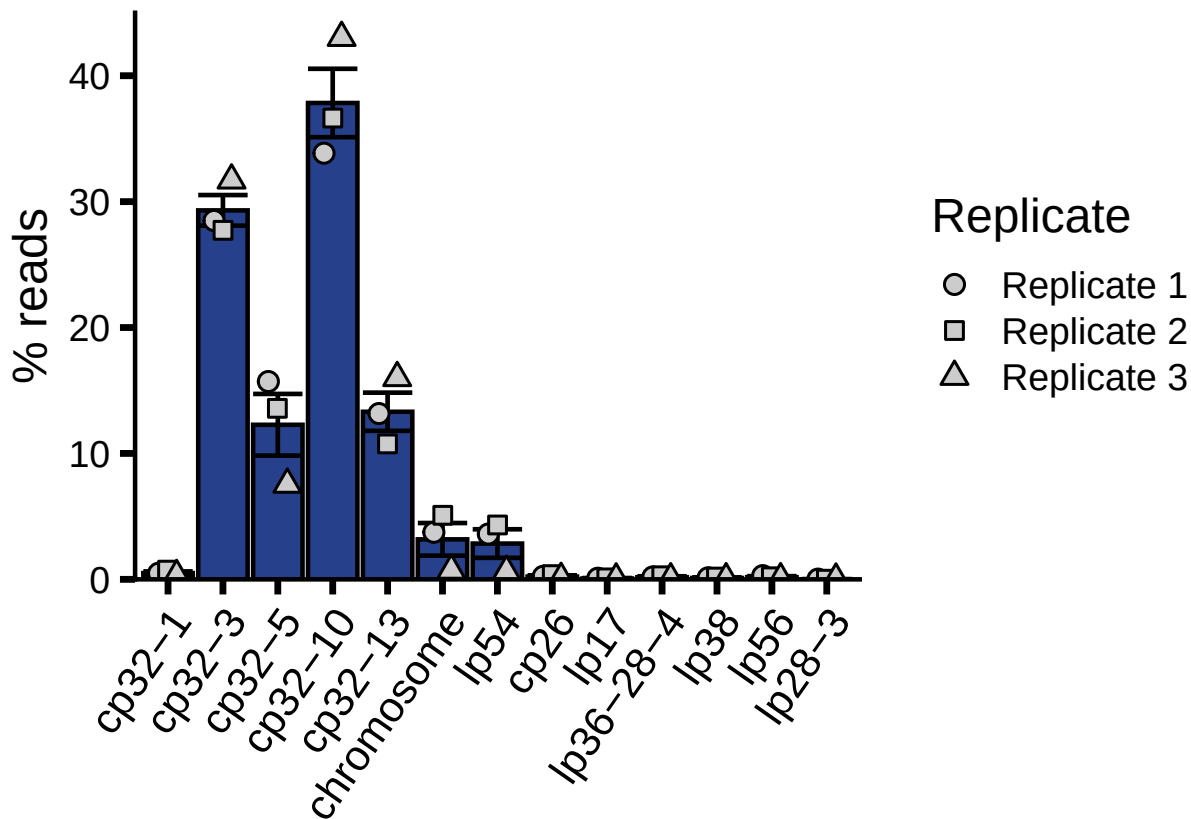
size = 3,
fill = "gray80",
color = "black",
stroke = 0.8
) +
scale_shape_manual(
  values = c(
    "Replicate 1" = 21,
    "Replicate 2" = 22,
    "Replicate 3" = 24 #manually include replicate shapes
  )
)

```


Original Published Figure 5B



Independent Reconstruction of Figure 5B



```

#installed ggbreak here. Removed for rmd
#install.packages ("ggbreak")
library(ggbreak)

## ggbreak v0.1.7 Learn more at https://yulab-smu.top/

## If you use ggbreak in published research, please cite the following
## paper:
##
## S Xu, M Chen, T Feng, L Zhan, L Zhou, G Yu. Use ggbreak to effectively
## utilize plotting space to deal with large datasets and outliers.
## Frontiers in Genetics. 2021, 12:774846. doi: 10.3389/fgene.2021.774846

#same process as Figure 5B
figCf_summary <- cov_allf |>
  group_by(Replicon) |>
  summarize(
    mean_depth = mean(meandepth),
    se_depth = sd(meandepth)/sqrt(n()),
    .groups = "drop"
  ) |>
  mutate(Replicon = factor(Replicon, levels = replicon_order))

#scale_y_break used to replicate broken y axis in original figure
#Unable to fix x-axis offset due to it being an artifact of ggbreak

figC <- ggplot(data = figCf_summary)+
  aes(x = Replicon, y = mean_depth)+
  labs(x = NULL, y = "Mean Coverage")+
  coord_cartesian(clip = "off")+
  scale_y_continuous(expand = c(0,0),
                     breaks = c(0, 20, 75, 150, 1000, 3000, 5000),
                     limits = c(0, 5000))+
  scale_y_break(c(150, 900),
               scales = 0.3,
               space = 0.8)+
  theme_classic(base_size = 18)+
  theme(
    axis.line = element_line(linewidth = 0.8),
    axis.ticks = element_line(linewidth = 1.2),
    axis.ticks.length = unit(0.2, "cm"),
    axis.text.x = element_text(angle = 55, hjust =1, size = 16),
    axis.line.y.right = element_blank(),
    axis.ticks.y.right = element_blank(),
    axis.text.y.right = element_blank(), #removed a right-sided y-axis
    panel.spacing = unit(0, "lines")
  )+
  geom_col(
    fill = "royalblue4",
    color = "black",
    linewidth = 0.8
  ) +
  geom_hline(

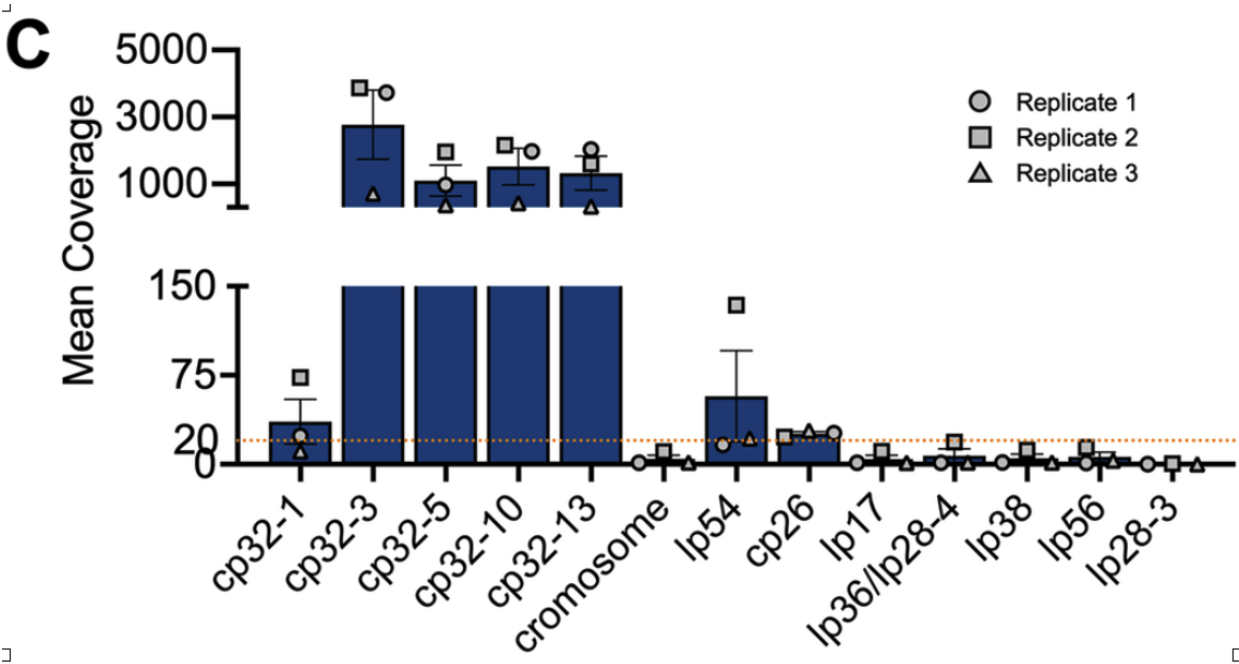
```

```

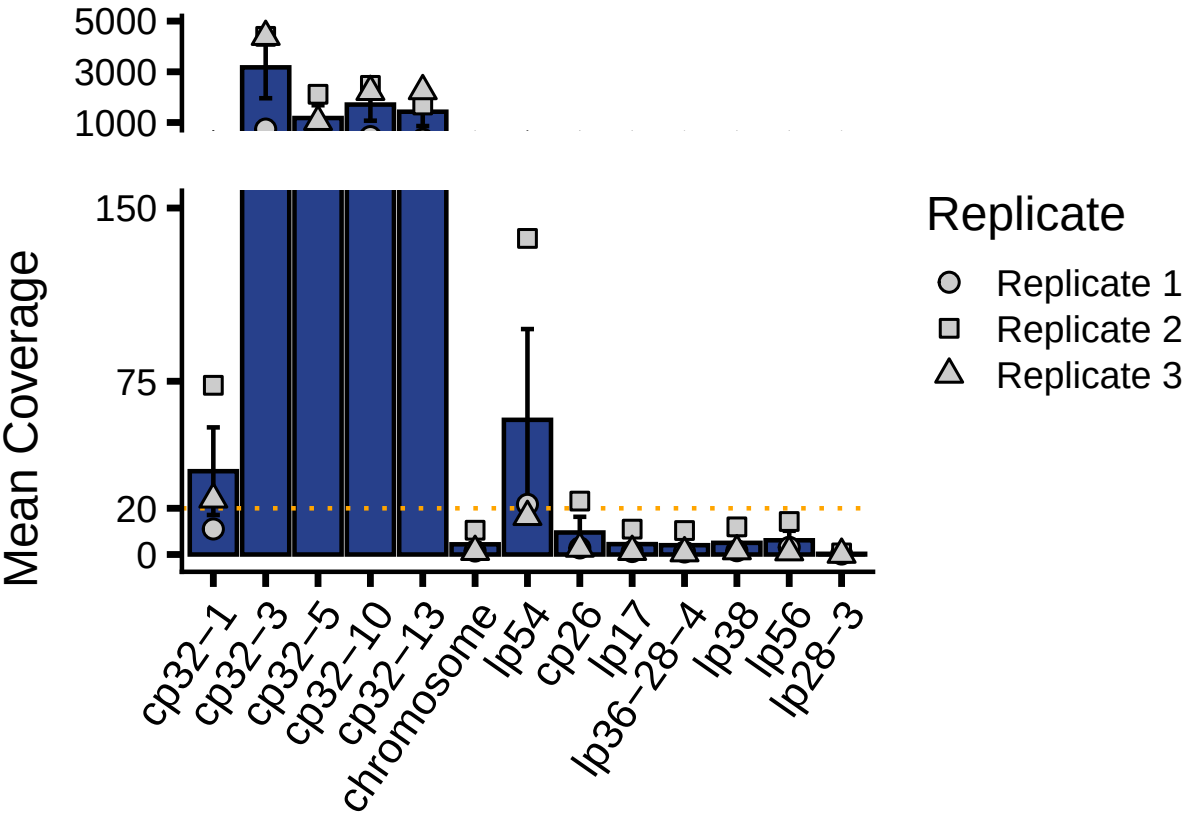
yintercept = 20,
linetype = "dotted",
color = "orange",
linewidth = 0.8
)+
geom_errorbar(
  aes(
    ymin = mean_depth - se_depth,
    ymax = mean_depth + se_depth),
  width = 0.25,
  linewidth = 0.8
)+
geom_point(
  data = cov_all_plot,
  aes(x = Replicon, y = meandepth, shape = Replicate),
  size = 3,
  fill = "gray80",
  color = "black",
  stroke = 0.8
) +
scale_shape_manual(
  values = c(
    "Replicate 1" = 21,
    "Replicate 2" = 22,
    "Replicate 3" = 24
  )
)

```

Original Published Figure 5C



Independent Reconstruction of Figure 5C



```

read_1 <- read_tsv("aln1_read_lengths.tsv")|>
  mutate(Replicate = "Replicate 1")
read_2 <- read_tsv("aln2_read_lengths.tsv")|>
  mutate(Replicate = "Replicate 2")
read_3 <- read_tsv("aln3_read_lengths.tsv")|>
  mutate(Replicate = "Replicate 3")
reads_all <- bind_rows(read_1,
                      read_2,
                      read_3)
reads_all <- left_join(reads_all, ref, by = "Reference")
reads_all <- reads_all |>
  mutate(Replicon = recode(Replicon,
                          "whole" = "chromosome"))
reads_all <- reads_all |>
  mutate(Replicon = factor(Replicon, levels = replicon_order))

#Preview of reads_all

reads_all|>
  head(17)|>
  kable(caption = "Raw Read Table 17/58k rows",
        booktabs=TRUE)

```

Table 8: Raw Read Table 17/58k rows

ReadID	Reference	ReadLength	Replicate	Replicon
SRR27421011.23738	NC_012157.1	8580	Replicate 1	cp26
SRR27421011.27725	NC_012157.1	6896	Replicate 1	cp26
SRR27421011.2261	NC_012157.1	9013	Replicate 1	cp26
SRR27421011.18161	NC_012157.1	6203	Replicate 1	cp26
SRR27421011.10159	NC_012157.1	6827	Replicate 1	cp26
SRR27421011.27330	NC_012157.1	5017	Replicate 1	cp26
SRR27421011.16227	NC_012157.1	5757	Replicate 1	cp26
SRR27421011.24115	NC_012157.1	8076	Replicate 1	cp26
SRR27421011.13281	NC_012157.1	6070	Replicate 1	cp26
SRR27421011.9560	NC_012157.1	5897	Replicate 1	cp26
SRR27421011.747	NC_012157.1	6109	Replicate 1	cp26
SRR27421011.18352	NC_012157.1	7352	Replicate 1	cp26
SRR27421011.24964	NC_012157.1	6842	Replicate 1	cp26
SRR27421011.23552	NC_012157.1	5221	Replicate 1	cp26
SRR27421011.1616	NC_012158.1	31446	Replicate 1	cp32-1
SRR27421011.1661	NC_012158.1	38254	Replicate 1	cp32-1
SRR27421011.2303	NC_012158.1	17868	Replicate 1	cp32-1

#No replicates, all three long-read runs were likely combined

```

figD <- ggplot(data = reads_all)+
  aes(x = Replicon, y = ReadLength)+
  coord_cartesian(xlim = NULL, ylim = c(0,100000))+
  scale_y_continuous(expand = 0)+
  labs(x = NULL, y = "Read Length")+
  theme_classic(base_size = 18)+

```

```

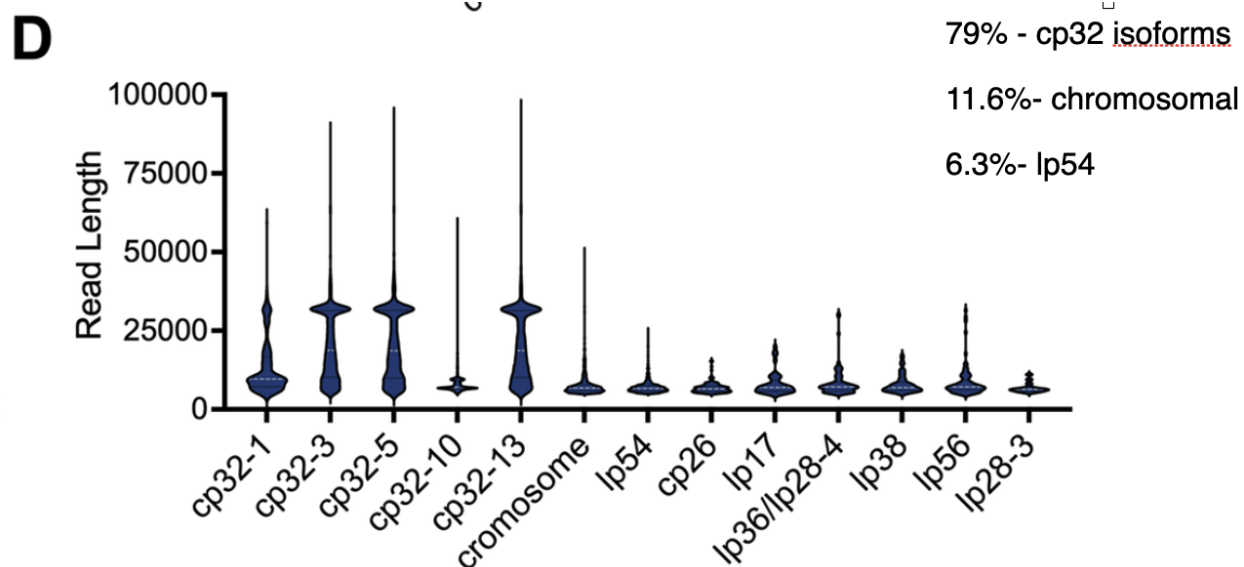
theme(
  axis.line = element_line(linewidth = 1.2),
  axis.ticks = element_line(linewidth = 1.2),
  axis.ticks.length = unit(0.2, "cm"),
  axis.text.x = element_text(angle = 55, hjust =1, size = 16))+
geom_violin(
  adjust = 0.7,
  scale = "width",
  trim = TRUE,
  width = 0.7,
  linewidth = 0.5,
  draw_quantiles = c(0.25, 0.5, 0.75),
  fill = "royalblue4",
  color = "black") #adjust, scale, trim, and width for improved visualization

```

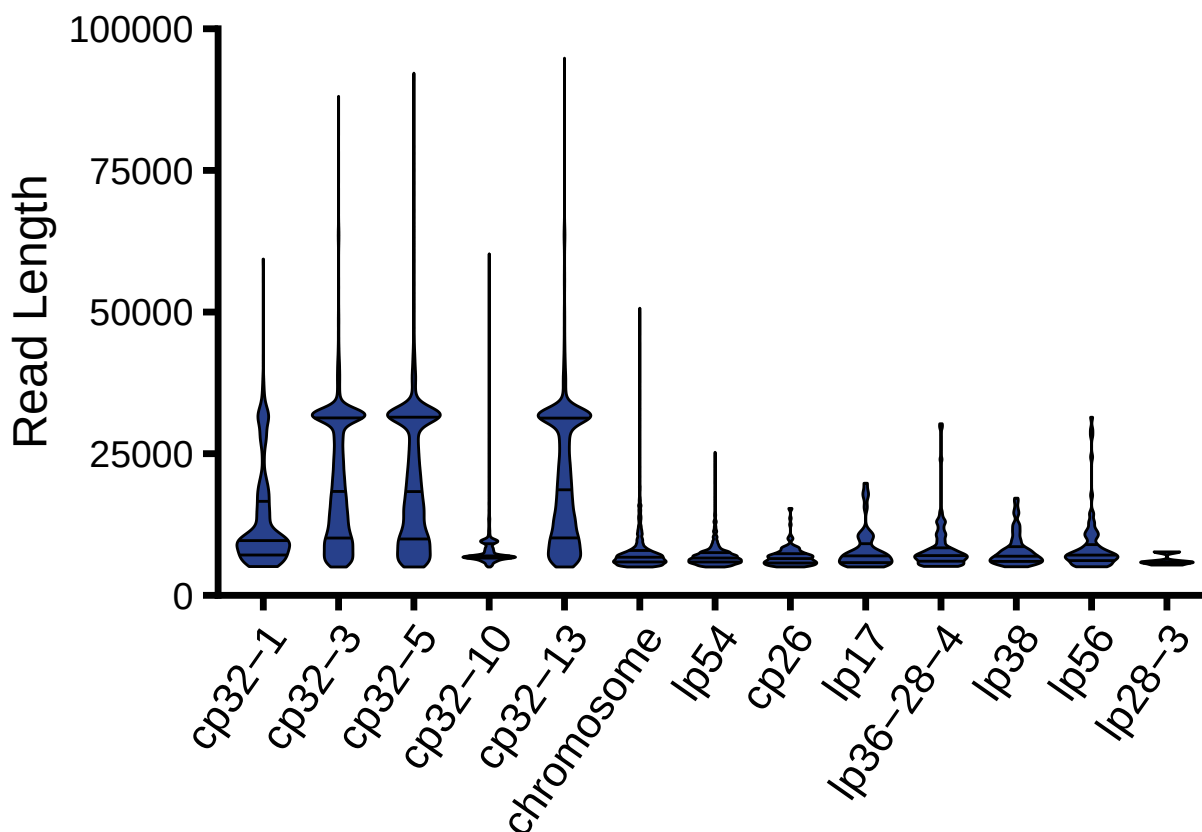
```

## Warning: The 'draw_quantiles' argument of 'geom_violin()' is deprecated as of ggplot2
## 4.0.0.
## i Please use the 'quantiles.linetype' argument instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

```



Independent Reconstruction of Figure 5D



Full Analysis of reconstructions can be found in the BIOL792.02_phiBB1_report. Generally, the reproduced figures seem to match the same trends observed in the original publication. Differences may be due to differences in methods and tools used at certain steps and a lack of access to the original source code.